

Parametrization of I - V characteristics of solar cells via asymptotics solutions of the super-ellipse model

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ABSTRACT

The computation of I - V characteristic curves is vitally important for tasks related to the design, sizing, and simulation of photovoltaic modules, as it allows them to evaluate the performance of different PV technologies under changing weather conditions. A simple and computationally efficient way to calculate I - V curves involves using the explicit super-ellipse model. However, this model is complicated to parameterize, as it requires numerically solving complicated systems of nonlinear equations. In response to this problem, this paper presents analytical methodologies for parameterizing the super-ellipse model. Two approaches are developed: one utilizing the maximum power point (MPP) and another employing two measured points from the I - V curve. The methods use asymptotic approximations to solve the equations, avoiding computationally intensive numerical techniques. The model is validated with experimental corresponding to six PV technologies and 18 weather and irradiance conditions using RMSE as the evaluation metric. The results show that the asymptotic solutions achieve performance comparable to numerical methods while offering simplicity and computational efficiency, presenting a RMSE of less than 0.0705 W for the methodology that uses the MPP and 0.027 A for the one that uses the measurement of two points on the I - V curve. In addition, it was verified that this model complies with the IEC EN 50530 standard. This study also shows the dependencies of Super-ellipse model parameters with irradiance and temperature. The proposed approaches address limitations of prior models, and enhance the representation of I - V curves under diverse operational scenarios.

Nomenclature

See Table 1.

1. Introduction

The characteristic curves of photovoltaic (PV) modules determine their electrical behavior under different irradiance and temperature conditions. The computation of current-voltage (I - V) curve is of central importance, as it allows engineers responsible for the design, control and optimization of solar farms to evaluate the performance of PV modules [1]. The single diode model (SDM) is one of the most widely used models and its parameters can be computed analytically using the analytical formulas presented by Batzelis in [2]. However, the computation of the I - V curve becomes complex due to the use of functions such as Lambert's W , which is commonly programmed in commercial math software, such as Maple, Matlab, or Mathematica. An alternative approach is to use the so-called explicit models, which

present mathematical expressions that directly relate the modules current to the voltage and its characteristic parameters. Various explicit models exist in the literature and have been studied and evaluated in detail by Pindado et al. in [3].

One of the most significant challenges for these types of model is determining their parameters so that the theoretical predictions align with the experimental measurements of the I - V curve accurately [4]. There are mainly two parametrization methodologies: (i) based on measurements of the I - V curve, and (ii) using physical parameters of the SDM. In the first case, it is necessary to have as many measurements as parameters, in order to formulate a system of equations which can be solved to obtain the value of the parameters. These points can be the open-circuit voltage (V_{oc}), the short-circuit current (I_{sc}), the maximum power point (MPP) or the measurement of some arbitrary point along the I - V curve. The second methodology consists in finding formulas that relate the parameters of the explicit model to the SDM. A summary of the numerous investigations related to this problem and the type of methodology used for its characterization is shown in Table 2.

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Table 1

Nomenclature for the equations describing the solar cell I - V characteristics, including symbols, their definitions, and respective units.

Symbol	Definition	Units
I	Output current of the solar cell	A
V	Output voltage of the solar cell	V
P	Output power of the solar cell	W
I_L	Photogenerated current	A
I_0	Recombination current	A
R_s	Series resistance	Ω
R_p	Shunt resistance	Ω
q	Elementary charge	eV
A	Diode ideality factor	Dimensionless
N_s	Number of cells in series	Dimensionless
V_{th}	Thermovoltage	V
k_B	Boltzmann constant	eV/K
T	Temperature	K
ξ & η	Super-Ellipse parameters	Dimensionless
I_{mpp}	Current at maximum power point	A
I_{sc}	Short-circuit current	A
i	Current at maximum power point normalized	Dimensionless
V_{mpp}	Voltage at maximum power point	V
V_{oc}	Open-circuit voltage	V
v	Voltage at the maximum power point normalized	Dimensionless
(V_a, I_a)	A measurement of the I - V curve to a percentage a of V_{oc} .	(V,A)
(V_b, I_b)	A measurement of the I - V curve to a percentage b of V_{oc} .	(V,A)
APE	Absolute Percentage Error	%
RMSE	Root Mean Square Error	–

Table 2

Summary of explicit models with the methodologies used to compute their parameters.

Model	Methodology.		SDM
	I - V curve points	MPP	
	I - V Measurements		
Akbaba & Alattawi (1995)	[5]	[3]	–
El-Tayyan (2006)	–	[3,6]	–
Karmalkar Haneefa (2008)	[7,8]	[3,9]	[7]
Das & Saetre (2011)	[10]	[3,11,12]	[13]
Das (2013)	–	[3]	[14]
Pindado & Cubas (2017)	[3]	[15]	–
Dash (2015)	[8]	–	–
Miseli (2015)	–	[16]	–

One of the simplest approaches is the so-called super-ellipse model, which takes the idea of relating the characteristic equations of an ellipse with the I - V curve. Initially, this was presented by Saetre et al. in [11] using the MPP to compute the parameters and independently by Das in [10] using two points of the I - V curve. Recently, the multivariate Newton–Raphson method was applied by Olayiwola et al. in [12] to estimate parameters with the MPP. In [13] and [17], Das et al. established that the super-ellipse model can be viewed as an approximation of the SDM and the double diode model, respectively. In [18], Das also studied the inverse case, where formulas were presented to compute the value of the SDM parameters based on the parameters of the super-ellipse models. Nevertheless, the analytical formulas presented so far lack accuracy, while the numerical methods can be computationally expensive and inherently have divergence problems.

Although explicit methods solve the problem of efficiently calculating the I - V curve, the calculation of its parameters becomes a complex task, as it requires numerically solving a system of nonlinear equations. Given the lack of exact analytical expressions that are simple and easy to implement in code, this work presents formulas for computation of the characteristic parameters of the super-ellipse model. The analytical approaches are derived using two methodologies: (i) by means of the measurement of MPP and (ii) with two points of the I - V curve. A nonlinear multi-Variable equation system, similar to the one presented

by Olayiwola et al. in [12], is formulated, which in turn reduces to a single-variable transcendental equation. It is found that under specific conditions, it is possible to apply the analytical methodology presented by Olver in [19,20] to find higher order asymptotic approximations to estimate the parameters. Using expressions relating the MPP to the SDM parameters presented by Tirado in [21], the dependence of the parameters of the super-ellipse model on different conditions of irradiance and temperature is evaluated. The formulas are evaluated using different technologies, irradiances, and temperature conditions, under experimental measurements performed by NREL [22]. The model presents a RMSE smaller than 0.0705 W and 0.027 A for each methodology and complies with the IEC EN 50530 standard.

This article is organized as follows: in Section 2, the preliminary mathematics, a brief problem description, and the two parametrization methodologies proposed are presented. In Section 3, the results of the model applied to six technologies and several weather conditions. Finally, Section 4 presents the conclusions of this article.

2. Mathematical methods

2.1. Preliminary mathematics

This section presents the mathematical tools used to asymptotically estimate the super-ellipse model parameters. Let $x \in \mathbf{R}$ be a variable satisfying

$$x = u + \phi(x), \quad (1)$$

where $u \in \mathbf{R}$ is a constant and $\phi(x)$ is a function that satisfies

$$\lim_{x \rightarrow \infty} \phi(x) = 0. \quad (2)$$

Eq. (2) is equivalent to writing $\phi(x) = o(1)$, where $o(\cdot)$ denotes Landau's little-o symbol, a notation commonly used to describe the asymptotic behavior of functions. In Landau's little-o notation, $\phi(x) = o(1)$ means that $\phi(x)$ becomes negligibly small compared to 1 as $x \rightarrow \infty$ [20]. More generally, if there exists a function $\xi(x)$ such that

$$\lim_{x \rightarrow \infty} \frac{\phi(x)}{\xi(x)} = 0, \quad (3)$$

then, one may write

$$\phi(x) = o(\xi(x)), \quad (4)$$

indicating that $\phi(x)$ is asymptotically smaller than $\xi(x)$.

In the limit, $x \rightarrow \infty$, $\phi(x)$ in (1) vanishes. Then, the solution of (1) is approximated by

$$x = u + o(1) \quad (u \rightarrow \infty). \quad (5)$$

The function $\phi(x)$, called a *correction factor*, can be included in the approximation by making successive substitutions of (5) into (1) and expanding $\phi(x)$ in a power series [19]. If the initial approximation is not accurate enough, it is possible to perform re-substitutions indefinitely until the desired accuracy is achieved.

2.2. Problem description

The current I in the SDM follows

$$I = I_L - I_0 \left(\exp(\alpha(IR_s + V)) - 1 \right) - \frac{IR_s + V}{R_p}, \quad (6)$$

where: I_L is the photocurrent, I_0 is the saturation current, $\alpha = \frac{1}{AN_s V_{th}}$, A is the ideality factor of the diode, N_s is number of cells connected in series and V_{th} refers to the thermovoltage, R_p is the parallel resistance value and R_s is the series resistance. To compute of I - V curves using (6), it is necessary to solve for I using a numerical method [23] or to use the Lambert W function [24]. However, these methods are complex to use and may have divergence problems. A way to avoid these problems is to use explicit models, such as the *Super-Ellipse* model,

presented initially in [11] and independently in [10]. The $I-V$ curves in this model are found using

$$I = I_{sc} \left(1 - \left(\frac{V}{V_{oc}} \right)^\eta \right)^{\frac{1}{\xi}}, \quad (7)$$

from where it is possible to solve for V to obtain

$$V = V_{oc} \left(1 - \left(\frac{I}{I_{sc}} \right)^\xi \right)^{\frac{1}{\eta}}, \quad (8)$$

where V_{oc} is the open-circuit voltage, I_{sc} is the short-circuit current, ξ and η are the model's parameters. The computation of the $I-V$ curve becomes simpler using (7) or (8) instead of (6).

2.3. Parametrization through MPP

The power is given by the product $P = IV$. At the MPP the following, it satisfied that

$$\frac{dP}{dV} = \frac{dP}{dI} = 0. \quad (9)$$

From (9), it follows that the current and voltage at the MPP is given by

$$I_{mpp} = I_{sc} \left(\frac{\eta}{\eta + \xi} \right)^{\frac{1}{\xi}}, \quad (10)$$

$$V_{mpp} = V_{oc} \left(\frac{\xi}{\eta + \xi} \right)^{\frac{1}{\eta}}. \quad (11)$$

If one has access to reliable estimations of I_{sc} , V_{oc} , I_{mpp} and V_{mpp} , it is possible to compute the values of the parameters ξ and η employing (10) and (11). To do this, rewrite first (10) as

$$\left(\frac{I_{mpp}}{I_{sc}} \right)^\xi - \frac{\eta}{\eta + \xi} = 0. \quad (12)$$

Then, clearing ξ from (11) gives

$$\xi = \frac{\eta}{\left(\frac{V_{oc}}{V_{mpp}} \right)^\eta - 1}, \quad (13)$$

which can be used to evaluate to Eq. (12) to obtain an expression only in η as

$$\left(\frac{I_{mpp}}{I_{sc}} \right) \left(\frac{V_{oc}}{V_{mpp}} \right)^{\frac{\eta}{\eta + \xi}} - \frac{\eta}{\eta + \frac{\eta}{\left(\frac{V_{oc}}{V_{mpp}} \right)^\eta - 1}} = 0. \quad (14)$$

For the computation η it is necessary solve (14) numerically.

2.3.1. Asymptotical solution

The method presented in Section 2.1 can be used to find an asymptotical solution of the transcendental problem. For this, it is necessary to represent (14) in the form of (1). To simplify the analysis, let the dimensionless parameters v , and i be given by

$$v := \frac{V_{mpp}}{V_{oc}}, \quad (15)$$

$$i := \frac{I_{mpp}}{I_{sc}}. \quad (16)$$

Eq. (14) is then expressed as

$$i^{\frac{\eta}{v^{-\eta}-1}} = \frac{\eta}{\eta + \frac{\eta}{v^{-\eta}-1}}. \quad (17)$$

Algebraic manipulations in (17) yield

$$i^{\frac{\eta}{v^{-\eta}-1}} = \frac{\eta}{\left(1 + \frac{1}{v^{-\eta}-1} \right) \eta}$$

$$i^{\frac{\eta}{1-v^{-\eta}}} = 1 + \frac{1}{v^{-\eta}-1}$$

Table 3

Minimum and maximum values of the characteristic parameters η and ξ from the literature.

Reference	η		ξ	
	Min	Max	Min	Max
[3]	1.964	32.421	0.612	1.959
[4]	1.996	16.599	0.515	2.144
[10]	5.300	17.000	0.320	1.140
[11]	12.000	14.000	0.700	0.900
[12]	3.808	16.651	0.517	2.084
[17]	3.149	19.306	0.334	2.024

$$\begin{aligned} &= 1 + \frac{1}{\frac{1-v^\eta}{v^\eta}} \\ &= 1 + \frac{v^\eta}{1-v^\eta} \\ &= \frac{1-v^\eta+v^\eta}{1-v^\eta} \\ &= (1-v^\eta)^{-1}. \end{aligned}$$

Applied the natural logarithm to both sides gives

$$\frac{\eta}{1-v^{-\eta}} \ln(i) = -\ln(1-v^\eta). \quad (18)$$

Let $a := -1/\ln(i)$. Then (18) is expressed as

$$\eta = a(1-v^{-\eta}) \ln(1-v^\eta). \quad (19)$$

According to the values presented in Table 3 from the literature, $\eta > 0$, and furthermore $\eta \gg \xi$, implying that η takes large values [3, 4, 10–12, 17]. To find the asymptotical behavior of (19), the following limit is considered

$$\begin{aligned} \lim_{\eta \rightarrow \infty} a(1-v^{-\eta}) \ln(1-v^\eta) &= a \lim_{\eta \rightarrow \infty} (1-v^{-\eta}) \ln(1-v^\eta) \\ &= a \lim_{\eta \rightarrow \infty} \ln(1-v^\eta) - a \lim_{\eta \rightarrow \infty} \{v^{-\eta} \ln(1-v^\eta)\}. \end{aligned}$$

Since $0 < v < 1$, the first limit on the right-hand side is

$$\lim_{\eta \rightarrow \infty} \ln(1-v^\eta) = 0, \quad (20)$$

which, in Landau's little-o notation, is equivalent to writing $a \ln(1-v^\eta) = o(1)$ when $\eta \rightarrow \infty$. Using L'Hôpital's rule for the second limit yields

$$\lim_{\eta \rightarrow \infty} v^{-\eta} \ln(1-v^\eta) = -1. \quad (21)$$

Therefore

$$\lim_{\eta \rightarrow \infty} \eta = \lim_{\eta \rightarrow \infty} a(1-v^{-\eta}) \ln(1-v^\eta) = a. \quad (22)$$

Then, in the limit $\eta \rightarrow \infty$, Eq. (19) becomes in

$$\eta = a + o(1), \quad (\eta \rightarrow \infty), \quad (23)$$

which, according to [19], is equivalent to

$$\eta = a + o(1), \quad (a \rightarrow \infty). \quad (24)$$

Note that (24) follows the same form as (5). Then, to incorporate the dependence with v , the right-hand of (19) is expanded in power series for large η values as follows

$$(1-v^{-\eta}) \ln(1-v^\eta) = 1 - \frac{v^\eta}{2} - \frac{v^{2\eta}}{6} + o(v^{3\eta}), \quad (25)$$

Substituting (25) into (19) results in

$$\eta = a - a \frac{v^\eta}{2} - a \frac{v^{2\eta}}{6} + o(v^{3\eta}). \quad (26)$$

Note that (26) has the same structure as (1). Here the variable x is η ; the constant u is a , and the function $\phi(\eta)$ equals to $\phi(\eta) = -av^\eta/2 - av^{2\eta}/6 + o(v^{3\eta})$, because $\lim_{\eta \rightarrow \infty} \phi(\eta) = o(1)$. Substituting (24) into (26), results in the first and second order approximations

$$\eta = a \left(1 - \frac{1}{2} v^a \right) + o(v^a), \quad (27)$$

$$\eta = a \left(1 - \frac{1}{2} v^{a(1-\frac{1}{2}v^a)} - \frac{1}{6} v^{2a(1-\frac{1}{2}v^a)} \right) + o(v^{2a}), \quad (28)$$

where (27) and (28) include the dependence with v explicitly in the computation of η . Obtaining higher order terms in the approximation becomes more and more complex, therefore for the present work, the series is truncated to second order. The computation of ξ is carried out by substituting (24), (27) or (28) into (13) depending on the accuracy required.

2.4. Parametrization through two-point measurement of the $I-V$ curve

The second approach for computing the parameters η and ξ takes two measurements of the $I-V$ curve at points $(V_a, I_a) = (aV_{oc}, j_a I_{sc})$ and $(V_b, I_b) = (bV_{oc}, j_b I_{sc})$, with $0 < a, b < 1$. Employing these measurements, it is possible to form a system of simultaneous equations using (7) and (8) as follows

$$a^\eta + j_a^\xi = 1, \quad (29)$$

$$b^\eta + j_b^\xi = 1. \quad (30)$$

Isolating ξ from both expression leads to

$$\xi = \frac{\log(1 - a^\eta)}{\log(j_a)}, \quad (31)$$

$$\xi = \frac{\log(1 - b^\eta)}{\log(j_b)}. \quad (32)$$

Equating both equations

$$\frac{\log(1 - b^\eta)}{\log(j_b)} = \frac{\log(1 - a^\eta)}{\log(j_a)}. \quad (33)$$

Then, multiplying both side of (33) by a^η/b^η results into

$$\frac{a^\eta \log(1 - b^\eta)}{b^\eta \log(j_b)} = \frac{a^\eta \log(1 - a^\eta)}{b^\eta \log(j_a)}, \quad (34)$$

which is equivalent to

$$\frac{a^\eta}{b^\eta} = \frac{a^\eta \log(1 - b^\eta)}{b^\eta \log(j_b)} \frac{\log(j_a)}{\log(1 - a^\eta)}. \quad (35)$$

Applying the logarithm in both sides yields

$$\log\left(\frac{a^\eta}{b^\eta}\right) = \log\left(\frac{a^\eta \log(j_a)}{b^\eta \log(j_b)} \frac{\log(1 - b^\eta)}{\log(1 - a^\eta)}\right). \quad (36)$$

And solving for η gives

$$\eta = \frac{1}{\log\left(\frac{a}{b}\right)} \log\left(\frac{\log(j_a)}{\log(j_b)} \left(\frac{a}{b}\right)^\eta \frac{\log(1 - b^\eta)}{\log(1 - a^\eta)}\right). \quad (37)$$

Note that (37) has the same form as (1). This can be seen by rewriting

$$\eta = \bar{\eta} + g(\eta). \quad (38)$$

where

$$\bar{\eta} = \frac{\log\left\{\frac{\log(j_a)}{\log(j_b)}\right\}}{\log\left(\frac{a}{b}\right)}, \quad (39)$$

$$g(\eta) = \frac{\log\left\{\left(\frac{a}{b}\right)^\eta \frac{\log(1 - b^\eta)}{\log(1 - a^\eta)}\right\}}{\log\left(\frac{a}{b}\right)}. \quad (40)$$

2.4.1. Asymptotical solution

The method presented in Section 2.1 can be used to find a solution to Eq. (38). Analyzing the function $g(\eta)$ when η is large, i.e., calculating the limit

$$\lim_{\eta \rightarrow \infty} g(\eta) = \frac{1}{\log\left(\frac{a}{b}\right)} \lim_{\eta \rightarrow \infty} \left\{ \log\left(\frac{\log(1 - b^\eta)}{\log(1 - a^\eta)}\right) \right\}. \quad (41)$$

Table 4

Expressions for computing the parameters from prior literature.

Ref.	Parameters	
	ξ	η
[3]	$-\frac{a^\eta}{\log(i)}$	$-\frac{1}{\log(i)}$
[10]	$-\frac{a^\eta}{\log(j_a)}$	$\frac{\log(\log(j_a)/\log(j_b))}{\log(a/b)}$

Using (21) in the argument of logarithm yields

$$\lim_{\eta \rightarrow \infty} g(\eta) = \frac{1}{\log\left(\frac{a}{b}\right)} \lim_{\eta \rightarrow \infty} \left\{ \log\left(\frac{-1}{-1}\right) \right\} = 0, \quad (42)$$

In virtue of (3), it follows that

$$g(\eta) = o(1), \quad (43)$$

which satisfies (5). Thus, the solution of (38) is approximated by

$$\eta = \bar{\eta} + o(1). \quad (44)$$

To improve accuracy, Eq. (44) may be used to evaluate (38) recursively. Expanding in power series of large η to first order approximation yields

$$\eta = \bar{\eta} + \frac{b^{\bar{\eta}} - a^{\bar{\eta}}}{2 \log\left(\frac{a}{b}\right)} + o(b^{\bar{\eta}} - a^{\bar{\eta}}), \quad (45)$$

and to second order approximation yields

$$\begin{aligned} \eta = \bar{\eta} + \frac{1}{\log\left(\frac{a}{b}\right)} & \left(\frac{1}{2} \left[b^{\bar{\eta} + \frac{b^{\bar{\eta}} - a^{\bar{\eta}}}{2 \log\left(\frac{a}{b}\right)}} - a^{\bar{\eta} + \frac{b^{\bar{\eta}} - a^{\bar{\eta}}}{2 \log\left(\frac{a}{b}\right)}} \right] \right. \\ & \left. + \frac{5}{24} \left[b^{2\bar{\eta} + \frac{b^{\bar{\eta}} - a^{\bar{\eta}}}{\log\left(\frac{a}{b}\right)}} - a^{2\bar{\eta} + \frac{b^{\bar{\eta}} - a^{\bar{\eta}}}{\log\left(\frac{a}{b}\right)}} \right] \right) \\ & + o(b^{2\bar{\eta}} - a^{2\bar{\eta}}). \end{aligned} \quad (46)$$

Finally, the computation of ξ is performed by substituting (44), (45) or (46) into (31) or (32). In [3,10], the expressions shown in (4) are presented. Note that those expressions are equivalent to Eqs. (24) and (44). Therefore, equations from Table 4 can be seen as special cases of the asymptotic approximation presented in this paper.

3. Results

This section addresses the validation of the formulas for the computation of the η and ξ parameters using the methodologies described in this paper. The approximations were validated using six different PV technologies and eighteen different weather conditions.

3.1. Using $I-V$ curve measurements for different technologies

Table 5 shows the PV metrics for six different technologies recorded in Golden Colorado USA by the National Renewable Energy Laboratory (NREL) and presented in [22,25]. Fig. 1 shows six $I-V$ measurements and those reproduced using the explicit model. The tags for each type of technology are

- “aSiMicro03036” for Amorphous microcrystalline silicon
- “mSi0188” and “mSi460A8” for Multi-crystalline silicon
- “CIGS1-001” for Copper indium gallium selenide
- “xSi12922” for Single-crystalline silicon
- “CdTe75669” for Cadmium telluride
- “HIT05662” for heterojunction with intrinsic thin-layer

In order to evaluate the performance of the super-ellipse model for reproducing the $I-V$ curves two criteria are used. The first criterion uses the IEC EN 50530 standard, which states that the absolute error

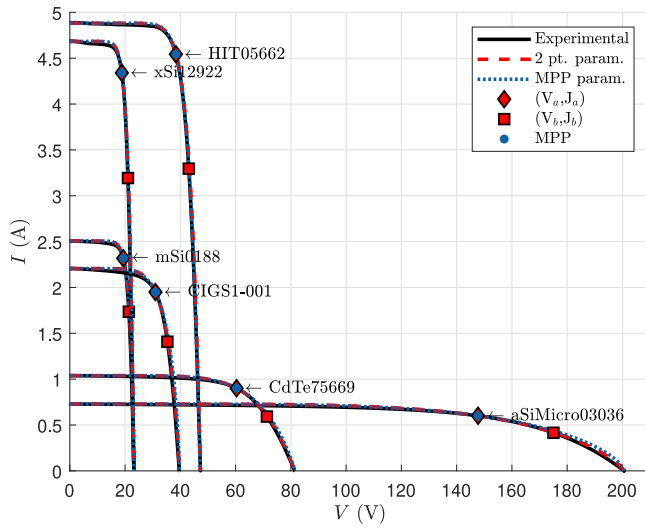


Fig. 1. Experimental (black) and theoretical I - V curves generated by the super ellipse model using the MPP measurement (dashed blue) and two points of the IV curve (dashed red) for six different technologies. Marking (V_a, I_a) with red diamonds and (V_b, I_b) with squares, while the MPP with a blue dot. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 5

PV metrics and (V_b, I_b) point for six technologies presented in [22].

Module	I_{sc} (A)	V_{oc} (V)	I_{mpp} (A)	V_{mpp} (V)	V_b (V)	I_b (A)
aSiMicro03036	0.729	200.58	0.601	147.74	175.03	0.416
mSi0188	2.507	23.191	2.319	19.337	21.37	1.736
CIGS1-001	2.206	39.615	1.95	30.991	35.315	1.409
xSi12922	4.683	23.153	4.34	18.833	21.043	3.193
CdTe75669	1.038	81.445	0.901	60.359	71.332	0.592
HIT05662	4.882	47.258	4.543	38.39	43.057	3.296

between the model and experimental measurement should be less than or equal to 1%, in the vicinity of $\pm 10\%$ of I - V and power-voltage (P - V) curve around the MPP. Initially, it was defined mathematically in [26], but recently, a detailed formulation was presented in [4]. This criterion is useful for evaluating the performance of parametrization technique that uses MPP measurement. The errors ϵ_I for the current and ϵ_P for power are given by

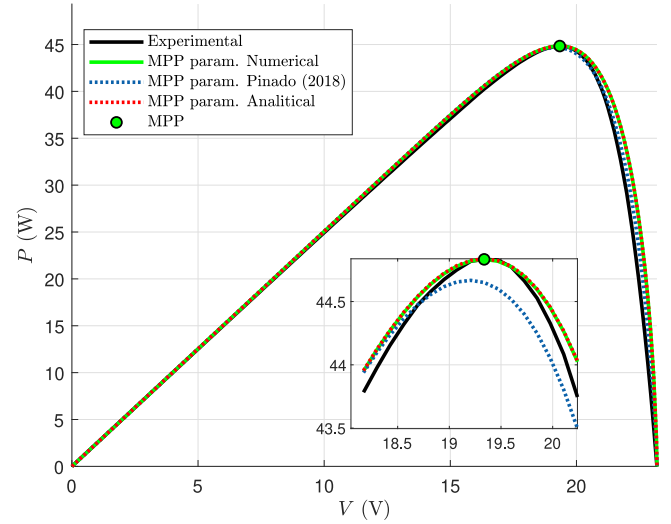
$$\epsilon_I = \frac{1}{0.2V_{mpp}} \int_{V_{mpp} \pm 10\%} \left| \frac{I_{model}(V) - I_{exp}(V)}{I_{exp}(V)} \right| dV \times 100, \quad (47)$$

$$\epsilon_P = \frac{1}{0.2V_{mpp}} \int_{V_{mpp} \pm 10\%} \left| \frac{P_{model}(V) - P_{exp}(V)}{P_{exp}(V)} \right| dV \times 100, \quad (48)$$

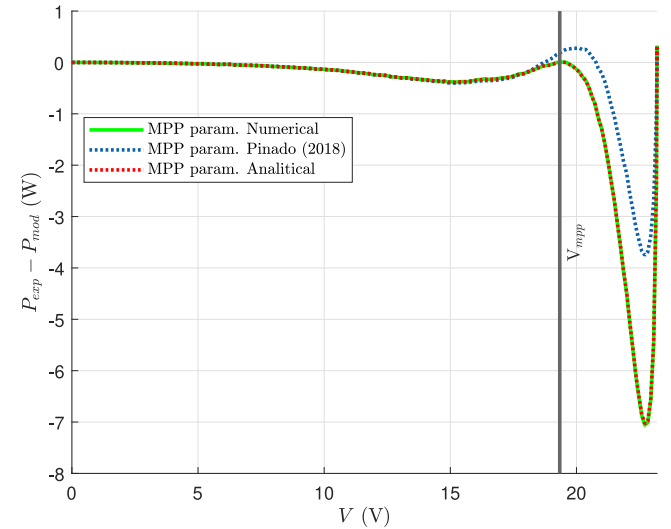
where $I_{exp}(V)$ refers to the experimental I - V curve, $P_{exp}(V)$ is the experimental P - V curve, $I_{model}(V)$ is the I - V curve computed using (7) and $P_{model}(V) = I_{model}V$. The integrals in (47) and (48) are computed using the trapezoidal rule. The second criterion is the *Root Mean Square Error* (RMSE), which is given by

$$RMSE = \sqrt{\frac{1}{N} \sum_{k=1}^N (I_{exp} - I_{model})^2}. \quad (49)$$

Table 6 shows the RMSE for the model with parameters computed using the first presented approach. The second and third columns of Table 6 present the ξ and η parameters computed using the formulas presented in [3] (see Table 4). The sixth and seventh columns shown the parameters estimate by (28) and (13). The 10th and 11th columns display the numerically computed parameters using (19). The Matlab function *fsolve* with (28) as an initial guess was employed to compute the numerical solution. It is observed that in five out of six modules,



(a)



(b)

Fig. 2. Experimental P - V curve (black), the calculated using the formulas presented in [3] and shown in Table 4 (blue dashed), analytical expression (28) (red dashed) and numerical solution of (19) (green) from mSi0188 module (a), followed by plot of power error between the experimental measurements and theoretical estimations (b), for each methodology. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

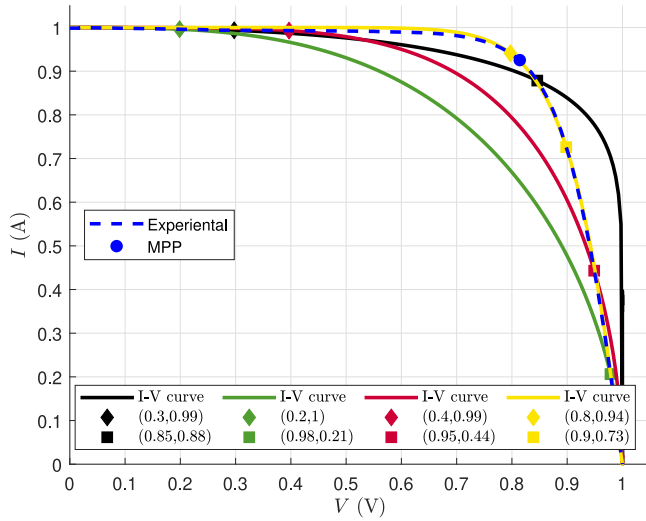
the approximation presented in [3] shows a lower RMSE compared to the analytical and numerical approximations. This implies that the model presented in [3] has a higher accuracy for representing the I - V curves in this case. On the other hand, in five out of six cases, the numerical solution (19) presents better performance considering ϵ_P . Fig. 2(a) shows the power-voltage (P - V) curve for the mSi0188 module using the methodology that employ the MPP measurement. Fig. 2(b) shows the error between the experimental and simulated P - V curves. It is noted that expression (28) shows very similar performance to numerical method. In the neighborhood of MPP, the model proposed coincides exactly with the experimental measurements. It should be noted that for voltage values between V_{mpp} and V_{oc} , the formulas of [3] show higher accuracy.

Table 6

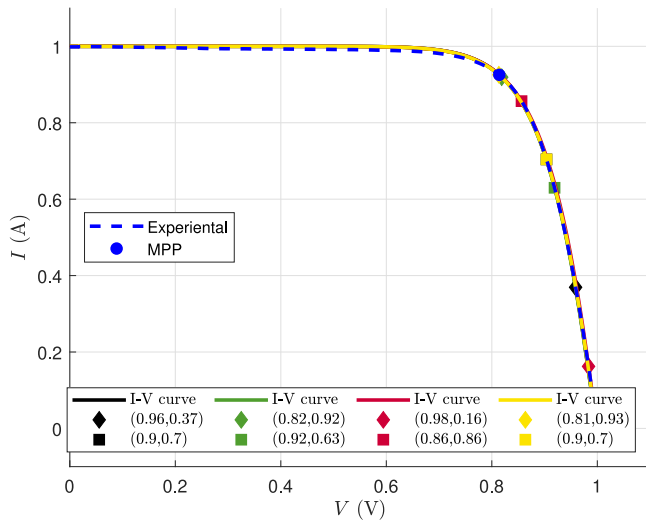
Parameters ξ and η computed with MPP using the formulas presented in [3], asymptotical approximation of (28) and numerical solution of (19). Followed by the RMSE and ε_1 for each case.

Module	η	ξ	RMSE (W)	ε_1	η	ξ	RMSE (W)	ε_1	η	ξ	RMSE (W)	ε_1
	Pindado [3]				Asymptotical				Numerical			
aSiMicro03036	5.18	1.060	0.0182 ^a	2.278	4.50	1.521	0.0182 ^a	0.361	4.45	1.536	0.0183	0.199 ^a
mSi0188	12.80	1.250	0.0606 ^a	0.412 ^a	12.10	1.516	0.0701	0.770	12.05	1.519	0.0704	0.764
CIGS1-001	8.10	1.110	0.0575 ^a	0.934	7.43	1.430	0.0655	0.523	7.39	1.437	0.0659	0.488 ^a
xSi12922	13.20	0.869	0.0403 ^a	0.405	12.70	0.998	0.0533	0.316	12.67	0.998	0.0535	0.314 ^a
CdTe75669	7.05	0.852	0.0159 ^a	0.868	6.55	1.071	0.0182	0.310	6.53	1.075	0.0183	0.282 ^a
HIT05662	13.9	0.773	0.0222	0.478	13.50	0.871	0.0165 ^a	0.135	13.48	0.872	0.0165 ^a	0.135 ^a

^a Minimum value



(a)

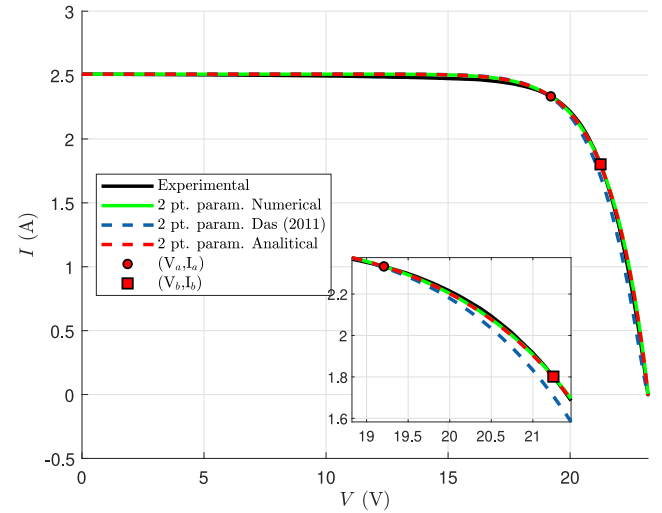


(b)

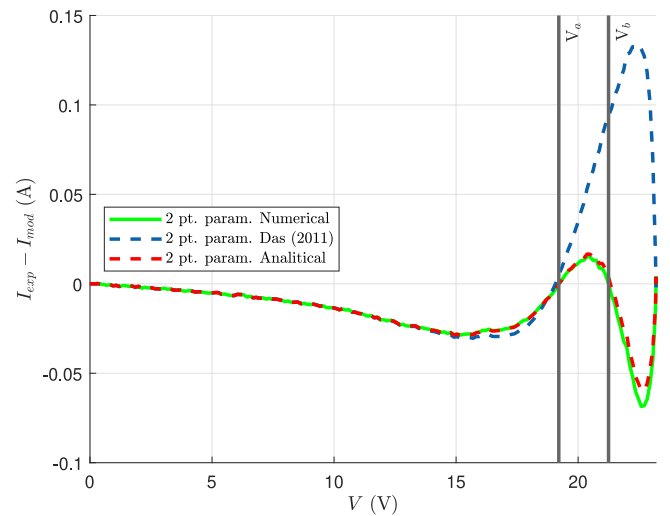
Fig. 3. I - V curve from mSi0188 module and the estimation of super-ellipse module using a and b values between zero and one (a), and around V_{oc} (b).

3.2. Choice of a and b

The choice of measurement points along the I - V curve impacts strongly on the performance of this methodology. Two arbitrary values are suggested in [10]: $a = 0.8$ and $b = 0.9$. According to (44), (45) and (46), the present model requires only that $a \neq b$, since otherwise the term $1/\log\left(\frac{a}{b}\right)$ would not be defined. Fig. 3(a) shows the normalized



(a)



(b)

Fig. 4. Experimental I - V curve (black), the calculated using the formulas presented in [10] (blue dashed) and shown in Table 4, analytical expression (46) (red dashed) and numerical solution of (19) (green) from mSi0188 module (a), followed by plot of power error between the experimental measurements and theoretical estimations (b), for each methodology. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

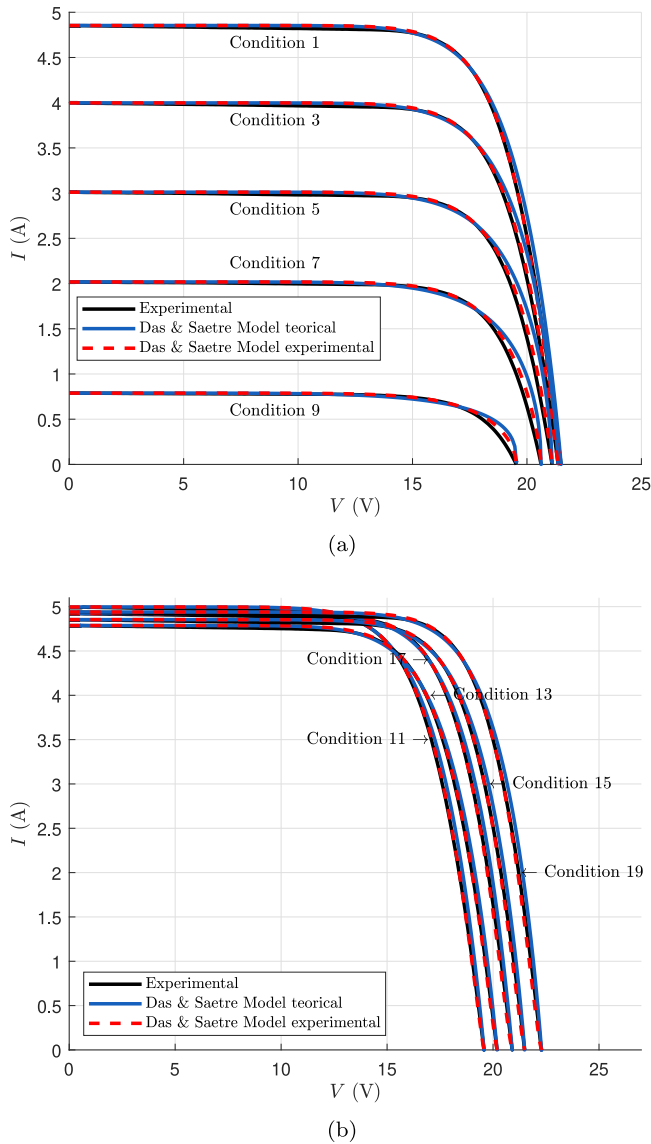


Fig. 5. I - V curves for mSi460A8 module taking into account: temperature constant and varying irradiance (a) using the conditions 1, 3, 5, 7, and 9, and irradiance constant and varying the temperature (b) using 11, 13, 15, 17 and 19 condition from Table 10.

I - V curves for different values of a and b , using data from the mSi0188 module. It is seen that the model satisfies its objective of fitting the designated points; however, it shows poor performance for the remaining points in the curve. Another constraint that the models must satisfy is that the I - V curve must coincide at the MPP. For this reason, in the case where $a = 0.8$ and $b = 0.9$, the model fits the experimental curve, since the normalized MPP usually lies in those ranges [27,28]. The present work suggests that a and b take the values $a = V_{mpp}/V_{oc}$ and $b = (1+a)/2$, i.e., the mid-point between V_{mpp} and V_{oc} . In this way, it is ensured that the V_{mpp} will lie in the generated I - V curve. Table 7 shows the RMSE between the model and the experimental curve for different values of a and b , as well as the estimates of the parameters ξ and η . It is observed that this strategy presents the lowest RMSE value. In case where an accurate MPP measurement is not available, one may choose a voltage value close to V_{oc} without compromising accuracy, as shown in Fig. 3(b) and in the last four rows of Table 7, where random voltage values between V_{mpp} and V_{oc} were selected. Note that the resulting values of a and b obtain a comparable RMSE.

Table 8 presents the parameters η , ξ , RMSE and ϵ_1 , computed using the formulas presented in [10] and shown in Table 4, (46) and the

Table 7

Estimation of η and ξ for different values of a and b using the I - V curve from the mSi0188 module, followed by the RMSE for each case.

η	ξ	a	b	RMSE (A)
2.437	8.440	0.300	0.850	0.13648
3.097	1.726	0.200	0.980	0.13431
4.678	1.874	0.400	0.950	0.06718
12.984	0.891	0.800	0.900	0.00597
13.438	0.843	0.963	0.905	0.00611
13.432	0.838	0.819	0.921	0.00621
13.231	0.888	0.986	0.858	0.00671
13.283	0.859	0.814 ^a	0.907	0.00594

^a MPP value

numerical solution of (38). In three of the six modules, the asymptotic solution has the lowest RMSE value and in the other three cases, it is the numerical solution that has the lowest RMSE. It is observed that the numerical solution has the lowest ϵ_1 , hence highest accuracy in reproducing the I - V curve in all cases. In both methods, it is noted that the value of (38) are very close to the numerical computation. It is also observed that in all studied cases, the methodology presented complies with the IEC EN 50530 standard, having values of $\epsilon_1 < 1$. Fig. 4(a) presents the I - V curve with an area magnification around the points (V_a, I_a) and (V_b, I_b) . It is observed that the numerical and analytical solution coincides with the two at the measurement points, unlike the formulas of [10] that coincide only in (V_a, I_a) . This increase in accuracy is due to the use of higher-order approximations compared to those in [10] which are of zero order. Fig. 4(b) shows the error between the experimental measurements and the models. Here, it is found that, in the interval from $(0, V_a)$ and (V_b, V_{oc}) , the models presented produce an overestimation of the current.

Table 9 compares execution times of the asymptotic method and the numerical approach (with $\eta_0 = 10$), in milliseconds, corresponding to the six technologies analyzed. As observed in columns two, four, seven, and nine, the asymptotic method estimates the parameters in shorter time, implying that it is more efficient in computational terms, especially the approach based on the measurement of MPP, when considering the average time of each estimation. One strategy to reduce the computational cost of the numerical method is to assign an initial condition as close as possible to the exact root. This not only decreases the computation time, but also the number of iterations required. Columns five and ten report the execution times of the numerical method using as an initial guess the approximation obtained by the asymptotic method, both for the two-point and the MPP-based methodology. In most cases, substantial reductions in execution times can be observed. Finally, columns three, six, eight, and eleven show the number of iterations required by the numerical methods. The results indicate a significant decrease in the number of iterations, which reflects an improvement in the convergence of the numerical methodology.

3.3. Using I - V measurements for different weather conditions

This section focuses in studying how ξ and η vary with respect to different irradiance and temperature conditions. Table 10 shows the data for 19 different weather conditions of the mSi460A8 module recorded in the city of Eugene, Oregon, USA. The first ten rows of Table 10 show the PV metrics (I_{sc} , V_{oc} , I_{mpp} and V_{mpp}) considering constant temperature, around of 25 °C, and variable irradiance, from 100 to 1000 (W/m²). The last nine rows show the PV metrics for constant irradiance around of 1000 (W/m²) and varying temperature from 15 to 55 (°C). To consider the dependence of the computation of ξ and η , the SDM its dependence on variable conditions are used. These were presented in [1], which considers the shunt resistance R_p , series R_s and diode ideality factor A as constants, and the recombination current I_0

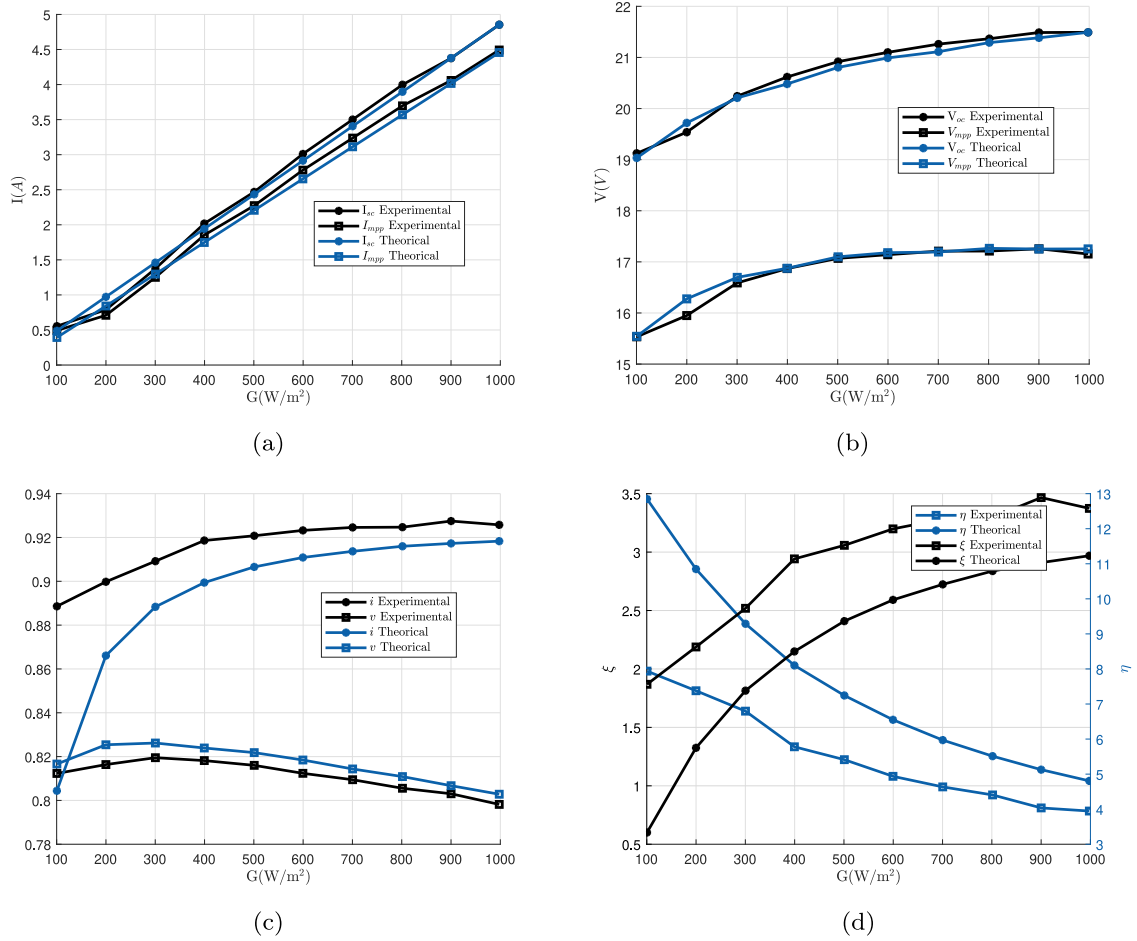


Fig. 6. Behavior of PV metrics of current (a) I_{sc} , I_{mpp} , voltage (b) V_{oc} , V_{mpp} , i and v ratios (c), ξ and η parameters (d), varying irradiance, considering theoretical estimations and experimental measurements.

Table 8

Parameters ξ and η computed with two points of I - V curve using the formulas presented in [3], asymptotical approximation of (28) and numerical solution of (19). Followed by the RMSE and ϵ_1 for each case.

Module	η	ξ	RMSE (A)	ϵ_1	η	ξ	RMSE (A)	ϵ_1	η	ξ	RMSE (A)	ϵ_1
	Das [10]				Asymptotical				Numerical			
aSiMicro03036	6.3	0.754	0.0298	2.33	5.22	1.173	0.0111 ^a	0.644	5.05	1.242	0.0113	^a 0.504
mSi0188	15.5	0.765	0.0459	1.55	14.13	1.020	0.0168 ^a	0.564	14.05	1.036	0.0177	^a 0.521
CIGS1-001	9.8	0.717	0.0606	1.78	8.75	1.005	0.0263 ^a	0.704	8.66	1.029	0.0269	^a 0.644
xSi12922	14.6	0.649	0.0943	1.04	13.49	0.837	0.0248	0.332	13.44	0.846	0.0240 ^a	^a 0.308
CdTe75669	8.2	0.597	0.0341	1.35	7.29	0.841	0.0109	0.411	7.21	0.863	0.0102 ^a	^a 0.358
HIT05662	14.8	0.642	0.1034	0.86	13.72	0.828	0.0213	0.207	13.67	0.837	0.0192 ^a	^a 0.184

^a Minimal value

Table 9

Execution time (in milliseconds) and iterations number for computing the I - V curves employ the two-point and MPP-based methodologies for six technologies using: (a) numerical, (b) Asymptotical and (c) numerical with asymptotical as initial guess.

Module	Two points					MPP				
	Num. (mS)	Iter.	Asym. (mS)	Num. & Asym. (mS)	Iter.	Num. (mS)	Iter.	Asym. (mS)	Num. & Asym. (mS)	Iter.
aSiMicro03036	131.89	10	1.2414	47.057	6	15.017	9	7.1320	11.523	4
mSi0188	39.076	7	1.8502	7.2817	5	14.249	7	2.2306	6.3413	4
CIGS1-001	5.1823	7	0.6729	3.9862	5	6.4356	8	0.5187	2.6409	4
xSi12922	3.2140	8	0.8132	1.9423	5	2.7101	8	0.4557	1.7812	4
CdTe75669	2.4240	11	0.0177	1.5175	8	1.6919	9	0.0127	1.3819	5
HIT05662	2.4421	8	0.0150	1.5370	5	1.4733	5	0.0101	1.1188	4
Mean	30.70	8.5	0.7684	10.550	5.666	6.9295	7.667	0.6568	4.1312	4.1667

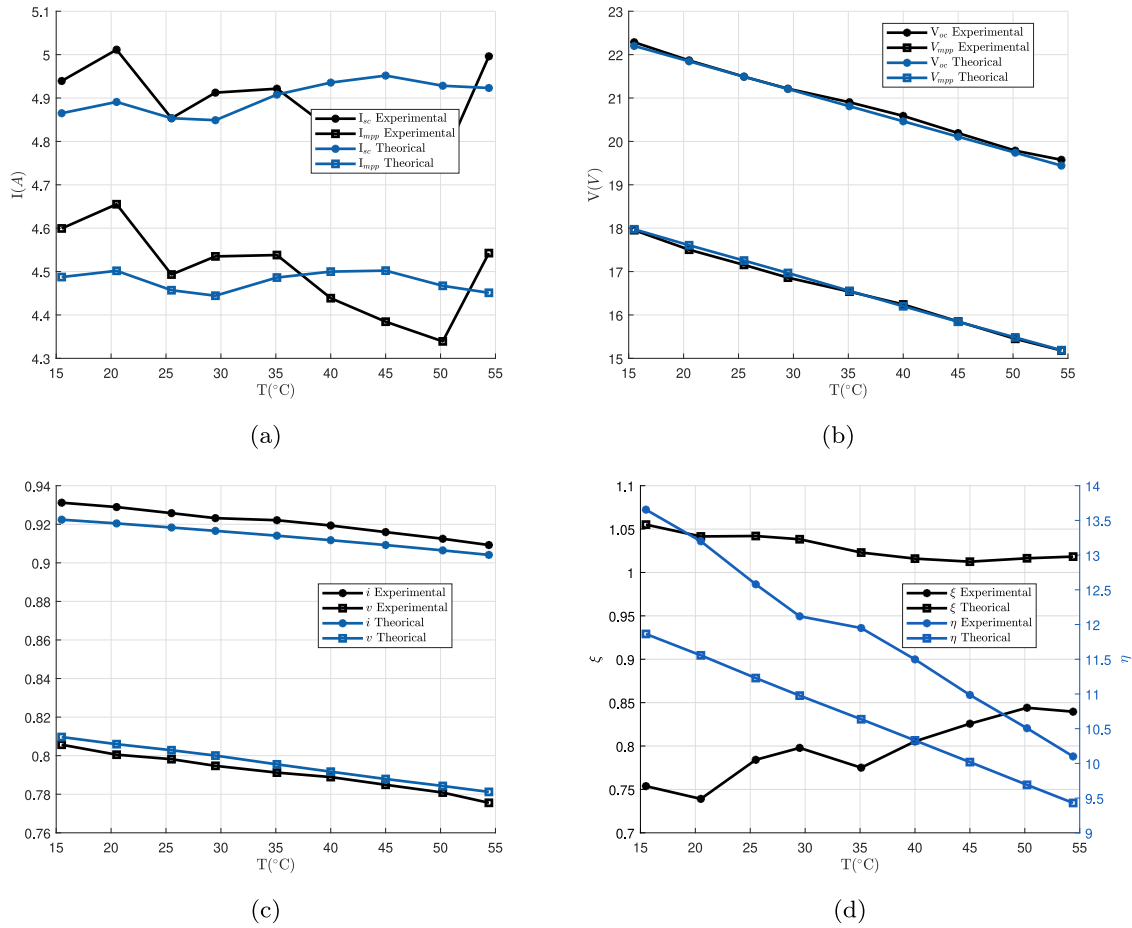


Fig. 7. Behavior of PV metrics of current (a) I_{sc} , I_{mpp} , voltage (b) V_{oc} , V_{mpp} , i and v ratios (c), ξ and η parameters (d), considering theoretical estimations and experimental measurements varying temperature.

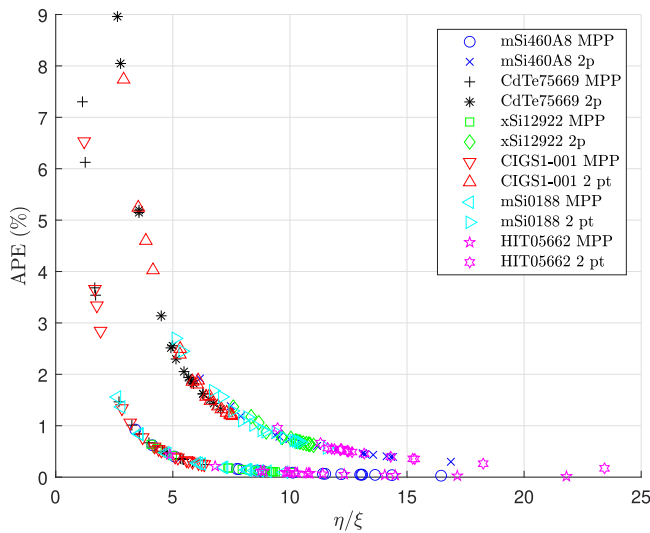


Fig. 8. APE between the numerical and asymptotic solutions as a function of the ratio η/ξ using the two-point and the MPP-based methodology for six different technologies under different temperature conditions ranging from 15 to 65 °C and from 100 to 1100 W/m² irradiance values provided by [25].

and photocurrent I_L being dependent on irradiance and temperature as follows

$$I_0 = \frac{I_{sc}^{STC} + \beta_{I_{sc}} \Delta T}{\exp \left\{ \frac{V_{oc}^{STC} + \beta_{V_{oc}} \Delta T}{n N_s V_{th}} \right\} - 1}, \quad (50)$$

$$I_L = \frac{G}{G_n} (I_{sc}^{STC} + \beta_{I_{sc}} \Delta T), \quad (51)$$

where V_{th} is the thermal voltage, $\beta_{I_{sc}}$ is the temperature coefficient of I_{sc} , $\beta_{V_{oc}}$ is the temperature coefficient of V_{oc} , I_{sc}^{STC} is the short circuit current and V_{oc}^{STC} is the open circuit voltage in standard condition testing (STC), i.e. $G_n = 1000$ (W/m²) and $T_n = 25$ °C, $\Delta T = T - T_n$, T and G is the temperature and irradiance incident on the module. The values of the SDM parameters at STC are computed using the analytical formulas present in [29] and are presented in the Table 11. The formulas for the computation of the PV metrics are given by [21,27]

$$V_{oc} = \log(I_L/I_0) A N_s V_{th} \quad (52)$$

$$I_{sc} = I_L \quad (53)$$

$$V_{mpp} = \frac{\ln(\mathcal{H}_V)}{\alpha \rho} - R_s (I_L + I_0) \quad (54)$$

$$I_{mpp} = I_L + I_0 - \frac{1}{\alpha R_p} \ln \left(\frac{\mathcal{H}_I}{I_0} \right) - \mathcal{H}_I, \quad (55)$$

where

$$\mathcal{H}_V = \frac{\exp \left(\frac{\rho r_{opt}}{r_{opt} - R_s} - 1 \right) ((\rho - 1) r_{opt} + R_s)}{\alpha \rho R_s I_0 (r_{opt} - R_s)}, \quad (56)$$

Table 10

PV metrics for different weather conditions, for constant temperature and varying irradiance (1–10 conditions) and holding irradiance constant and varying temperature (11–19 conditions), indicating the time at which the measurement recorded.

Cond.	Time	G (W/m ²)	T (°C)	I_{sc}	V_{oc}	I_{mpp}	V_{mpp}
1 ^a	2013-03-03T13:30:04	997.8	25.5	4.854	21.491	4.493	17.154
2	2013-11-21T10:45:04	900.1	25.4	4.376	21.488	4.059	17.256
3	2013-02-05T14:30:04	801.5	24.9	3.998	21.367	3.697	17.212
4	2013-11-27T14:10:04	700.4	25.3	3.501	21.261	3.237	17.210
5	2013-02-14T15:30:04	599.5	24.6	3.011	21.100	2.780	17.141
6	2013-03-24T14:20:04	500.8	24.4	2.469	20.918	2.274	17.069
7	2013-11-08T11:35:04	399.5	25.4	2.018	20.618	1.853	16.869
8	2013-06-16T07:50:04	300.3	24.8	1.375	20.243	1.250	16.589
9	2013-04-27T17:25:04	199.9	25.3	0.788	19.537	0.709	15.950
10	2013-08-29T10:30:04	100.5	24.3	0.555	19.123	0.493	15.534
11	2013-08-16T13:35:03	995.4	54.4	4.996	19.576	4.543	15.182
12	2013-08-05T12:35:04	997.3	50.2	4.755	19.786	4.339	15.451
13	2013-07-30T11:50:04	1001.4	45	4.787	20.192	4.385	15.849
14	2013-05-02T11:00:04	1001.4	40	4.828	20.588	4.439	16.242
15	2013-03-07T13:40:03	1000.2	35.1	4.922	20.903	4.538	16.539
16	2013-10-28T12:10:04	996.0	29.5	4.912	21.215	4.535	16.860
17	2013-03-03T13:30:04	997.8	25.5	4.8535	21.491	4.4932	17.154
18	2012-12-22T11:45:03	1003.3	20.5	5.011	21.866	4.655	17.504
19	2013-01-11T13:30:04	1002.3	15.5	4.939	22.285	4.599	17.955

^a STC data

Table 11

SDM parameters and temperature coefficients for the mSi460A8 module.

Parameters	Units
R_p	215.588 Ω
R_{p0}	215.853 Ω
R_s	0.2651 Ω
AN_s	43.213 –
I_0	1.927×10^{-8} A
I_L	4.856 A
β_I	3.225×10^{-3} A/°C
β_V	-7.088×10^{-2} V/°C

$$H_i = \left(\frac{R_p}{r_{opt} - R_s} - 1 \right) \frac{1}{\alpha R_p}. \quad (57)$$

In (56) and (57), the variables ρ and r_{opt} are introduced. This are given by

$$\rho = \frac{R_p}{R_s + R_p}, \quad (58)$$

$$r_{opt} = \frac{V_{mpp}^*}{I_{mpp}^*}, \quad (59)$$

with

$$V_{mpp}^* = \left(\frac{V_{oc} R_p}{(R_p + R_s)(V_{oc} - R_s I_{sc})} - 1 + \ln(G_V) \right) \cdot \frac{(R_p + R_s)}{\alpha R_p} - R_s(I_L + I_0), \quad (60)$$

$$I_{mpp}^* = I_L + I_0 - \left(\ln \left(\frac{1}{\alpha I_0 R_p} \left(\frac{I_{sc} R_p}{V_{oc} - I_{sc} R_s} - 1 \right) \right) + \frac{I_{sc} R_p}{V_{oc} - I_{sc} R_s} - 1 \right) \frac{1}{\alpha R_p}, \quad (61)$$

and

$$G_V = \left(\frac{V_{oc}}{(V_{oc} - R_s I_{sc})} - \left(1 + \frac{R_s}{R_p} \right) \right) \frac{1}{\alpha R_s I_0}. \quad (62)$$

Fig. 5(a) show the experimentally measured I – V curves and those estimated using (52)–(55) for conditions 1, 3, 5, 7 and 9 with variable irradiance and constant temperature at 25 °C. Fig. 5(b) for condition 11, 13, 15, 17 and 19 conditions for variable temperature with irradiance constant (around 1000 W/m²), where an adequate fit of the (7) model is observed.

Fig. 6(a) shows the theoretical and experimental currents I_{sc} and I_{mpp} varying with respect to irradiance. Fig. 6(b) shows the variation of

V_{oc} and V_{mpp} with respect to irradiance. In both cases, it is observed that the theoretical estimates are in good agreement with the experimental measurements. The accuracy of this estimate depends largely on the methodology used to estimate the SDM parameters, on the formulas for computing the variation of the parameters with respect to climatological changes, and on the intrinsic errors of the instruments used for the experimental measurements. The error show may be due to the aforementioned aspects. In addition, it can be seen that the currents I_{sc} and I_{mpp} have a linear dependence with respect to irradiance while the voltages V_{oc} and V_{mpp} grow with a logarithmic rate as shown by the formulas (52) and (54). Fig. 6(c) shows i and v , where it is observed that for the case of i the theoretical estimation produces an underestimation while for v it is overestimated. But in both cases the shape of the I – V curve in Fig. 5(a) is maintained. In the case of v , it is observed that there is a local maximum around $G = 300$ W/m², while i displays a monotonically increasing behavior. Fig. 6(d), the parameters ξ and η are shown varying the irradiance. It is observed that the theoretical estimate overestimates the value of ξ , while there is an underestimation of η . This discrepancy causes a slight error in the I – V curve for voltage values $V_{mpp} < V < V_{oc}$ as seen in Figs. 5(a) and 5(b).

Fig. 7 shows the case where the temperature is varying and the irradiance holds constant. While V_{oc} and V_{mpp} in Fig. 7(b) change linearly with temperature, the currents I_{sc} and I_{mpp} in Fig. 7(a) do not show a clear trend, it is due to a nonlinear dependence of the current with temperature of cell [30]. Nevertheless, the theoretical and measured i and v in Fig. 7(c) show a linear and monotonically decreasing dependence on temperature. Additionally, Fig. 7(d) shows a discrepancy between the estimated and experimental values of ξ and η . However, the I – V curve in 5(b) shows an adequate fit. In [17], it was stated that only v changes with respect to temperature and irradiance, while i remains constant. Figs. 6 and 7 clearly show that this parameter has a strong dependence on these changes, which is taken into account by the model presented in this work.

Fig. 8 presents a plot of the ratio η/ξ against absolute percent error (APE) between the numerical and the asymptotic solution. The η/ξ metric is chosen since the main assumption for obtaining the asymptotic estimates is that $\eta \gg \xi$. To obtain this graph, data from [25] was employed. These data corresponds to PV metrics reported under 18 different weather conditions, with temperatures varying between 15 and 65 °C and irradiances between 100 and 1100 W/m². The results show that, for values of the η/ξ ratio close to one, the APE increases rapidly, reaching a maximum of 8.96% for the methodology based on two points of the I – V curve, and 7.3% for the one based on MPP measurement, both cases corresponding to the “CdTe75669”

technology. This evidences that the main drawback of the model is that, when η/ξ is small, asymptotic approximations can yield significant errors. This behavior is not directly controllable, since the estimates depend on the measurements of the $I - V$ curves, and these vary between technologies and with weather conditions. On the other hand, the asymptotic behavior of the model is also visible, as the quotient η/ξ increases, the APE tends to zero, indicating an improvement in the performance of the estimates.

4. Conclusion

This work has introduced analytical solutions to parameterize the super-ellipse model for PV modules using asymptotic approximations. The use of asymptotic methods allows for finding analytical expressions to compute the parameters of model in a straightforward manner and with adequate accuracy. The presented approaches employ: (1) a measurement of MPP and, (2) two points on the IV curve. It is found that the two points can be anywhere within interval of (V_{mpp} , V_{oc}). The validation against experimental data from multiple PV technologies confirms the reliability of the derived formulas, with RMSE less than 0.071 W in reproducing $P-V$ curves for the approach that uses the MPP, and of less than 0.028 A in reproducing $I-V$ curves for the two-point methodology. The IEC EN 50530 standard is fulfilled in all cases too. The study of the behavior of the model parameters with respect to different weather conditions revealed that ξ and η have a nonlinear dependence with the irradiance and very close to linear dependence with the temperature. The asymptotic approach outperforms prior estimations in the most of cases, holding computational simplicity. The models presented provide users and engineers with an accurate and simple tool for estimating $I-V$ curves, supporting the design and optimization of solar energy systems. Future work will extend the methodology to dynamic weather conditions or partial shading.

CRedit authorship contribution statement

José G. Tirado-Serrato: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Alfredo Sanchez:** Writing – review & editing, Supervision, Project administration, Methodology, Formal analysis. **Serguei Maximov:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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